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The Motive to Borrow

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1. Introduction

The issuance of public debt is an important tool of economic policy. Borrowing can help governments deal with negative shocks, undertake countercyclical fiscal policy, and finance exceptionally large expenditures, such as public infrastructure investments. Many governments, particularly in advanced economies, responded to the Global Financial Crisis with exceptionally large debt-financed fiscal stimulus. In the United States, for instance, the Obama administration in 2009 approved the USD831 billion (5.5 percent of GDP) American Recovery and Reinvestment Act to help boost investment and job creation. US government debt increased from 64 percent of GDP in 2007 to above 100 percent in 2012. In advanced economies, the average debt-to-GDP ratio rose from about 60 percent in 2007 to over 90 percent in 2016. Stimulus spending and cyclically-lower revenues also resulted in higher public debt—at various levels of government—in many emerging markets as well. China is a case in point. The government embarked on a massive infrastructure and public investment program, spending more than 6 percent of GDP in discretionary stimulus measures and allowing public debt to increase from 29 percent of GDP in 2007 to 44 percent of GDP by 2016. More recently, the big infrastructure push associated with the China's Belt and Road Initiative is contributing to growing public debts and possibly to sustainability risks in some emerging market and developing countries (Case Study 3.1).

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Case Study 3.1 Balancing investment needs with debt sustainability, the case of Ethiopia

The 2030 Sustainable Development Agenda set ambitious targets for inclusive development, which will require a large scale up of investment over a long period of time. While the private sector should play a key role to mobilize resources, public investment is expected to increase significantly in several countries. At the same time, rising public debt is a source of concern in many developing countries, especially in Africa, where, after the sharp decline of debt levels thanks to the debt relief initiatives of the early 2000s, debt-to-GDP ratios are rising again—driven mostly by large primary deficits and the scaling-up of public investment (IMF 2018a). Rising debts and external imbalances are undermining debt sustainability and could pose a threat for future investment plans and sustained economic growth.

A case in point is Ethiopia. Public investment was above 7 percent of GDP in the 2000s and further accelerated in the last seven years (public investment was at about 15 percent of GDP between 2014 and 2017). This massive scale up of investment was funded by external concessional and non-concessional financing (including large Chinese investment flows), and partly facilitated by restrained government consumption, financial repression and an overvalued exchange rate (World Bank 2016). This policy mix allowed the expansion of a substantial physical infrastructure and the development of large projects, like the Grand Ethiopian Renaissance Dam and the railway connecting Addis Ababa with the port of Doraleh in Djibouti, which will significantly reduce trade costs and improve access to global markets for Ethiopian firms. The dam is estimated to cost almost USD 5 billion—about 5 percent of GDP—and once completed it will be the largest hydroelectric power plant in Africa, supplying energy also to Sudan and Egypt.

The large investments in infrastructure undertaken in recent years have started bearing fruits, as Ethiopia experienced a sustained rapid growth in the last decade, with real GDP growth averaging 10 percent annually. At the same time, poverty declined substantially and the provision of key public services has improved. The share of the population with access to electricity, for instance, has increased from 14 percent in 2005 to 43 percent in 2016.

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Case Study 3.1 *Continued*

Significant gaps still remain, however. According to estimates from the Global Infrastructure Hub, Ethiopia faces an investment shortfall of about USD 285 billion to achieve the targets set by the 2030 Agenda. But further investment has to be planned keeping in mind that the model used in the past to finance the infrastructure expansion is starting to show its limits in terms of debt sustainability, crowding out of private credit, and weak external competitiveness due to exchange rate appreciation (World Bank 2016). In particular, although the debt-to-GDP ratio declined from 107 percent in 2002 to 38 percent in 2009 (thanks to debt relief), since then it started to increase sharply and reached 62 percent in June 2018. Even in the presence of sustained economic growth, the adverse debt dynamics is reflected in the IMF and World Bank assessment, according to which Ethiopia is at high risk of debt distress (IMF 2018b). Moreover, absorptive capacity constraints could undermine the projects' success rate and reduce the dividend of public investment (Presbitero 2018). As a result, investment has recently started declining and the large external imbalances and the public debt burden are constraining future growth.

The experience of Ethiopia, although unique in a number of respects, can be generalized to other developing countries, at least with regard to the trade-off between investment financing and debt sustainability. A key lesson for policy makers is that, even in presence of strong growth and large investment needs, any investment scaling up has to consider the risks that debt-financed public investment and higher public debt could pose on debt sustainability and future economic growth.

While there are good reasons to issue debt, there are also political failures that induce governments to borrow too much—leading, in some cases, to public debt levels that are hard to rationalize as the optimal decision of a benevolent social planner. Such excessive debt accumulation may be costly inasmuch as it circumscribes future capacity to stabilize the business cycle or impairs economic growth, either by crowding out private investment or by increasing uncertainty about future tax and inflation rates. (Risks of public debt becoming unsustainable and leading to a debt crisis are discussed in Chapter 4.)

This chapter discusses why governments borrow, separating good reasons for issuing debt (Section 2) from bad ones (Section 3). Next, it describes the

link between public debt and economic growth (Section 4) before offering some concluding remarks (Section 5).

2. Good Motives to Borrow

Budget deficits are the buffer that governments use to delink intertemporally spending and revenues. Why not always balance the budget? The principle of tax smoothing states that during periods of exceptionally high spending the government should run a deficit to finance those expenditures with future taxes. These periods may be the result of various shocks (wars, natural disasters, etc.) or spending decisions that reflect an expected economic, financial, or social benefit. Increasing spending during recessions can smooth the business cycle—raising employment, output, and incomes. Public investment can lead to faster growth. Accommodating the need for structural reforms via short-run deficits can pay off in the future via higher GDP and higher tax revenues. In all these cases, the principle of tax smoothing states that the government should run a deficit. At the same time, the government must be mindful that the spending will indeed reap the expected benefit—and that it is not undertaking the expenditure simply because there is easy financing available.

In this section we begin by elaborating on the principle of tax smoothing and then discuss some of the cases where attention needs to be paid to the returns on investment. We finish by presenting an additional argument for the government to issue debt—namely to provide the private sector with a safe asset.

A. The logic of tax smoothing

Governments have long financed extraordinary expenditures by issuing debt—most notably, when fighting wars.[1] Often, it would have been socially and politically unacceptable to try to finance such a level of expenditure through contemporaneous taxation alone, so the government resorted to issuing debt.[2] But there is also a sound economic rationale. If the government can only raise distortionary taxes, and if the cost of the economic distortion is convex

[1] One of the few non-war related examples of "public" debt was the perpetual bonds issued by the water authority of Lekdijk Bovendams of the Netherlands in 1648. The Lekdijk Bovendams water authority, though not a sovereign government, had taxation powers over the residents protected by the dam; the bonds issued in 1648 continue to pay interest to this day.

[2] It is even possible that the government raises tax rates by so much that revenues *fall* (i.e., taxes are on the "wrong side" of the Laffer curve), making it physically impossible to finance the spending.

(i.e., increasing at an increasing pace) in the tax rate, then it makes sense to try to "smooth" taxes over time in order to minimize the total distortionary cost.

One of the first to articulate the concept of tax smoothing explicitly was US Secretary of the Treasury, Albert Gallatin (1807), who argued:

> It appears necessary to provide revenues at least equal to the annual expenses on a peace establishment, the interest on the existing debt, and the interest on loans that may be raised. [As to] whether taxes should be raised to a greater amount or loans be altogether relied upon for defraying the expenses of the war…the losses and privations caused by war should not be aggravated by taxes beyond what is strictly necessary. An addition to the debt is doubtless evil, but experience having now shown with what rapid progress the revenue of the Union increases in time of peace; with what facility the debt, formerly contracted, has been reduced; a hope may be confidently entertained that all the evils of war will be temporary and easily repaired; and that the return of peace will, without any effort, afford ample resources for reimbursing whatever may have been borrowed during the war.[3]

The idea was formalized by Barro (1979), who assumed a convex cost function, and showed that minimizing the present value of the distortionary burden involved equalizing the marginal cost of levying taxes over time. For a given tax base, this implies that the tax rate should be constant over time. If public expenditure is fixed, governments should run deficits and accumulate debt in bad times (when the tax base is cyclically low) and run surpluses and pay down debt in good times (when the tax base is high).[4]

While the logic of tax smoothing is clear, the problem that governments confront is that their expenditure is unlikely to be smooth—the classic example being that of a war. But the argument applies more generally: whenever there is some "lumpiness" in the government's spending, there will be a divergence between the time path of expenditure and the (optimally) constant taxes—with deficits (and hence debt) making up the difference. In fact, it can be shown that optimally, the government's overall balance should equal the present discounted value of expected future *changes* in government spending (Ghosh 1995a). Thus, tax smoothing applies only to temporary changes in spending: if there is a permanent increase or decrease in spending, then the

[3] See Hall and Sargent (2014).

[4] If taxes are not distortionary, Ricardian equivalence holds, and there are no transactions costs in the trading of government securities, then a form of the "Modigliani–Miller" theorem of public finance obtains—and the level of government debt at any moment is indeterminate (Barro 1979; Stiglitz 1988; Chan 1983). As discussed in section 2.D, however, there may still be good reasons for the government to issue debt—e.g., to provide a safe asset for financial markets.

expected change is zero, and the government should run neither a deficit nor a surplus. If the change in spending is permanent, then its burden cannot be spread over time to reduce the distortionary cost of the associated taxes. On the contrary, running a deficit in the face of a permanent rise in government spending would imply that—if the government is to respect its budget constraint (and not default)—eventually, it would have to raise taxes by even more to pay for both the higher spending and the interest on the accumulated debt thus violating the principle of tax smoothing.

Taking the logic one step further, if the government recognizes that it may face unexpected shocks (that raise spending above its normal level), it may want to "save for a rainy day" by accumulating assets or paying down debt (Aiyagari et al. 2002).[5] Calibration exercises for the US government by Bhandari et al. (2016) suggest that in the long-run, the government should hold a positive, albeit small, net asset position as precautionary savings against future spending shocks. In general, the government will want to hold a portfolio of debt and financial assets that minimizes the risk that it will have to alter tax rates across time or states of nature (Bohn 1990; Barro 1995). Building on this premise, numerous academic papers have explored the optimal capital structure of the government's assets and liabilities in a stochastic setting.[6]

Three points are noteworthy about the tax-smoothing argument for issuing debt. First, for a given the level of output (GDP), unless the government is borrowing from foreigners, the issuance of public debt does not increase the resource envelope of the economy (except when the fiscal stimulus raises output, as in the Keynesian models discussed in section 2.B below). Domestic public debt then necessarily crowds out private absorption (consumption or investment). Therefore, the only purpose of such borrowing would be to smooth taxes and thus lower the distortionary cost to the economy. But if the government borrows from foreigners—either directly or indirectly (i.e., issues debt to residents who in turn borrow from abroad)—then such borrowing would also expand the economy's real resource constraint, allowing the

[5] This idea was also anticipated by Gallatin, who in his 1807 Report wrote "A previous accumulation of treasure in time of peace, might, in a great degree, defray the extraordinary expenses of war, and diminish the necessity of either loans or taxes. It would provide during periods of prosperity, for those adverse events to which every nation is exposed, instead of increasing the burdens of the people when they are least able to bear them" (Gallatin 1807: 359). Again, this is in direct analogy to the intertemporal current account literature where uncertain national cash flow leads to the country running a larger surplus or smaller deficit than it would under certainty (Ghosh and Ostry 1997).

[6] See, for example, Kingston (1991), Zhu (1992), Chari et al. (1994), Barro (1995), Judd (1999), Angeletos (2002), Buera and Nicolini (2004), Marcet and Scott (2009). Berck and Lipow (2011) discuss how the tax-smoothing motive gets modified when the risk premium on government bonds is endogenous.

private sector to also smooth consumption against shocks to government spending. In such cases, there will be a positive association between the issuance of public debt and external (private or public) debt.

Second, while the government's ability to issue debt is welfare improving—inter alia, because it allows for tax smoothing—the debt itself becomes a dead-weight loss once issued. Even purely domestic debt—"a debt we owe ourselves"—represents an economic loss, equal to the present value of the economic distortions associated with the taxes necessary to repay it.[7] (If the public debt is held externally, then there is an additional real resource transfer that will need to be made to non-residents.) The greater the inherited debt, the higher the taxes required to service it. If taxes are distortionary and fall on any factor (e.g., labor or private capital) that is complementary to the productivity of public capital, then—optimally—a government that inherits a higher level of public debt will undertake less public investment, with corresponding effects on output and growth (Ostry et al. 2015).

Third, the logic of tax smoothing seems to apply regardless of the cost of borrowing. Intuitively, however, for a given spending shock, a government will want to borrow less the higher the interest rate it must pay on its debt. In fact, this result follows from the tax-smoothing argument. The simplest example is when output (the tax base) and government spending are constant. Taxes would then be set to equal government spending plus the interest on the stock of debt. In such a situation, if there was an unexpected, temporary increase in government spending (e.g., a war), then the government should deficit-finance the higher spending. Taxes would (by tax smoothing) be immediately and permanently raised to cover the additional interest on the debt that will have been accumulated during the period of exceptional spending. Now, if the government faces a high interest rate on its borrowing, then taxes have to be raised correspondingly high. But since taxes are raised immediately to their permanent higher level, for a given spending shock, a government that faces a high interest rate on its debt would run a smaller deficit (i.e., borrow less) than a government that faces a low interest rate.

[7] Even with purely domestic public debt, it makes a big distributional difference to who is the "we" and who is the "ourselves." In the Lucas and Stokey (1983) framework, the debt is held by a single representative agent but the government has only distortionary taxes at its disposal for servicing the debt. The optimal policy in this situation would obviously be for the government to default on its debt (since it is owed to the representative agent who also pays all taxes, but the act of servicing the debt imposes a deadweight distortionary cost on the economy). Since Lucas and Stokey rule out default by assumption, the time-consistent solution consists of a series of "mini-defaults" with each successive government manipulating the interest rate—by issuing more debt (which boosts current period consumption, and reduces the interest rate payable on the inherited debt).

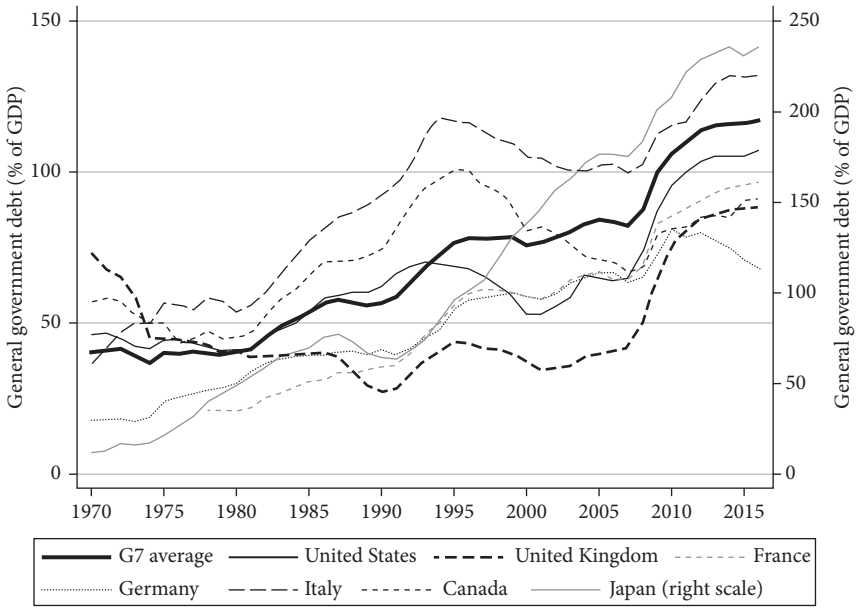

Figure 3.1 The evolution of the debt-to-GDP ratio in G7 countries
Source: Global Debt Dataset (Mbaye et al., 2018).

Summing up, the tax-smoothing argument suggests that countries should accumulate public debt to finance large and lumpy expenditures. While there is ample evidence that countries do accumulate debt during wars, tax smoothing is hard to reconcile with long-term debt accumulation during tranquil periods (the average debt-to-GDP ratio of the G7 countries rose from about 40 percent in 1970–80 to over 80 percent by 2007, see Figure 3.1). The link between debt accumulation and investment is even less clear. A simple regression that controls for country- and year-fixed effects shows a positive correlation between the debt-to-GDP ratio and public investment in advanced economies, implying that a 1 percentage point change in the debt-to-GDP ratio is associated with a 0.04 percentage point increase in public investment (Figure 3.2). This suggests that, typically, only a small percentage of debt issuance (4 percent) is used to finance public investment projects.[8] Bacchiocchi et al. (2011) find a negative correlation between debt and public investment in countries with a high debt ratio, and a positive correlation between debt and public investment in countries with low debt ratios. This may help explain the

[8] In emerging and developing economies, the correlation between public debt and public investment is instead negative but not statistically significant.

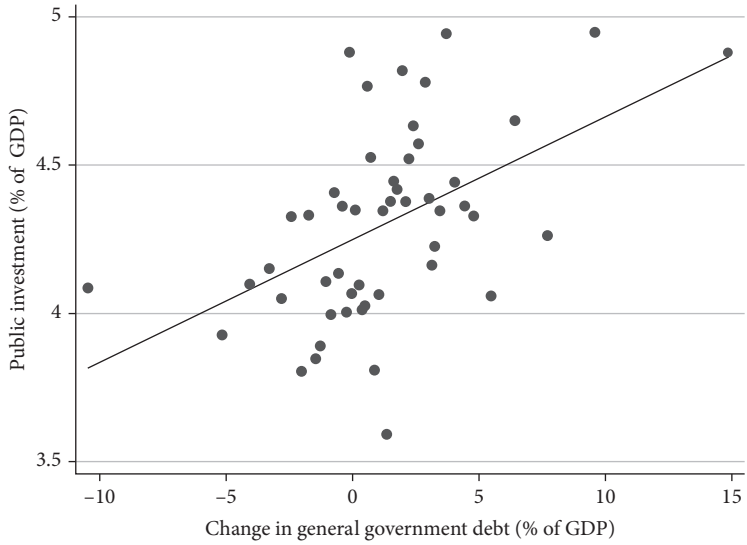

Figure 3.2 Correlation between change in public debt and contemporaneous public investment

Notes: A regression of the ratio of public investment over GDP at time *t* against the change in the ratio of general government debt over GDP between *t* and *t-1*, controlling for year and country fixed effects, gives a coefficient on the debt variable of 0.041 (p-value of 0.011), meaning that a 10% increase of the debt-to-GDP ratio is associated with 0.4% lower ratio of public investment over GDP. To generate the binned scatterplot, starting from the sample of nineteen OECD economies (data on general government for New Zealand are not available), the change in the ratio of general government debt over GDP between year *t* and *t-1* (x-axis) and public investment (as a % of GDP, y-axis) in year *t* are regressed against year and country fixed effects. Then, the x-residuals are grouped into fifty equal-sized bins and the chart plots, for each bin, the mean of public investment (as a % of GDP) in year *t*, within each bin, holding the controls constant. The diagonal line is the linear fit of the OLS regression of the y-residuals on the x-residuals. The number of observations is 899.

Source: Global Debt Dataset (Mbaye et al., 2018) and World Economic Outlook.

diversity of empirical results regarding the link between public debt and output growth, explored further in Section 4 below.

B. Keynesian demand stimulus

One example of temporary increases in spending is countercyclical fiscal policy that governments often implement during recessions. While in many ways this fits our previous logic of tax smoothing, it adds a second dimension and justification for that spending. The discussion thus far has taken output as given and considered optimal fiscal policy for an exogenous path of GDP. But during recessions, governments also try to boost output, so a full assessment of the benefits of running deficits must take account of this as well.

Why countercyclical policy is needed

In most macroeconomic models, monetary and fiscal policy are effective tools to stabilize the business cycle. In open economies, the Mundell–Fleming results imply monetary policy will be ineffective under fixed exchange rates and an open capital account, so only fiscal policy is available. The traditional Keynesian IS-LM model shows how changes in spending and taxes help stabilize aggregate demand by acting as a counteracting force to changes in private spending, which forms the basis of most policy discussions on the need for countercyclical policy (IMF 2008). More sophisticated (New Keynesian) models can also validate the IS-LM intuition in dynamic and optimizing environments (Beetsma and Jensen 2005).

In these models, monetary and fiscal policies are, in a sense, substitutes for stabilizing output. While monetary policy may be implemented faster and be less subject to political interference than fiscal policy—except in regard to automatic stabilizers, Taylor (2000)—there may be instances when monetary policy cannot achieve the first-best result, even with a flexible exchange rate.[9] For instance, monetary policy may be constrained by the zero lower bound on interest rates—as was the case for many central banks during the Global Financial Crisis—so the burden necessarily falls on fiscal policy (Eggertsson and Woodford 2004). More generally, in the presence of more than one distortion in the economy, not just price rigidity, monetary policy may not suffice to bring the economy to its first-best outcome and could be usefully complemented by fiscal policy (Blanchard and Gali 2010).

The normative statement that fiscal policy should be countercyclical is not always borne out empirically, however—especially in emerging markets, Latin American economies in particular (Gavin and Perotti 1997; Kaminsky et al. 2004). In part, this may be because the scope for deficit financing during downturns is more limited in such countries, while the lack of fiscal discipline during the upswing may reflect political economy considerations (as discussed in Section 3, below). Recent evidence—notably post-Global Financial Crisis—is more encouraging (Frankel et al. 2013).

The evidence for OECD or European economies is mixed. Fiscal policy is countercyclical but occasionally it turns procyclical, as in the case of recent fiscal consolidations in European countries (Égert 2012; Fatás 2018). One problem is judging the output gap in real time; especially after a financial

[9] When the government is not indifferent to the level of the exchange rate (or its external balance), monetary policy has traditionally been "assigned" to the external objective and fiscal policy to the internal objective (i.e., minimizing the output gap and maintaining full employment).

crisis, the trajectory for potential output—and hence the output gap—may have changed. Procyclical policy has negative economic consequences as it leads to higher output volatility and lower growth (Aghion et al. 2007).

Countercyclical policy, deficits, and debt

How do we characterize countercyclical fiscal policy? A commonly-used indicator of the fiscal policy stance is the change in the inflation-adjusted budget balance as a ratio to GDP (Blanchard 1993). Spending directly affects aggregate demand while taxes help stabilize disposable income and therefore private spending. The effects of taxes and spending might not be identical, but the budget balance comes close enough to capture their combined effect. This establishes a direct connection between stimulative fiscal policy and debt. When growth is below trend, governments will run deficits, and thus debt will accumulate.

It is important to distinguish between automatic and discretionary changes, even if from the perspective of aggregate demand this is largely irrelevant—it is the overall balance that matters. Automatic stabilizers capture changes in the budget balance that are the result of tax or spending laws that were not decided or modified as a result of current economic conditions. What is needed is not always strong cyclicality in taxes or spending. In fact, the largest source of automatic stabilizers in advanced economies is *acyclicality* of public spending. If the government maintains spending constant when GDP is falling, then even if taxes are proportional (so there is no extra countercyclicality in the tax schedule), deficits will increase. In this stylized case, the magnitude of the automatic stabilizers is simply proportional to size of government. The data show that the majority of automatic stabilizers among advanced economies comes from this effect (Fatás and Mihov 2012).

Beyond automatic stabilizers, governments also engage in discretionary fiscal policy changes. These changes follow the same logic as they also contribute to deficits and accumulation of debt during downturns. There is evidence, among advanced economies, that discretionary fiscal policy is used more aggressively in those countries that have the weakest automatic stabilizers (because of their smaller government size)—highlighting the substitutability between these two types of fiscal policy (Fatás 2009).

From cyclical deficits to accumulation of debt

Governments should—and do—run deficits during downturns. But if governments plan properly, then this should have no effect on public debt over the long run as debt would rise during downturns but decline during

recoveries.[10] As discussed in the section considering why countercyclical policy is needed, however, public debt was increasing in many advanced economies—despite the booming world economy—even before the Global Financial Crisis and the ensuing Great Recession.[11] Can this trend be related to the dynamics of stabilization policy over the cycle? Is there an asymmetry? And if so, why?

There are at least several possible sources of asymmetries. First, while governments are ready to apply countercyclical policies during recessions, they are less likely to follow the same logic during expansions. This is related to our earlier argument about observed procyclical policies (or not enough countercyclical policies). Empirically this is the case, at least for some countries (see Fatás and Mihov (2010) for a sample of European countries or Alesina et al. (2008) for a larger sample). The next section discusses the political distortions that may lead to such asymmetry.

The second reason is not so much about the political incentives of governments but about excessive optimism or pessimism when forecasting GDP growth. For example, during periods of strong growth, governments produce forecasts of potential output growth that are too optimistic. As the data confirm, estimates of potential output and its growth rate tend to be highly procyclical and this leads to excessive expansionary fiscal policy in good times (Mc Morrow et al. 2017). As an example, in December 28, 2000, President Clinton announced that the United States was on course to eliminate its government debt within the following 10 years. The macroeconomic scenario supporting this forecast did not incorporate the 2001 and 2008 recessions (nor, in fairness, did it foresee the Afghanistan and Iraq Wars).[12]

Is this bias in growth forecasts only present in good times? Not quite; we also observe excessive pessimism during downturns when potential output estimates are revised downwards. This leads to excessively tight fiscal policy during recessions, which in principle should offset the effect on debt of the excessive optimism during expansions. Beyond the difficulties in predicting turning points and lags in the implementation of discretionary fiscal measures (which can result in a net bias toward larger deficits and higher debt on

[10] This was part of UK Chancellor Gordon Brown's "Golden Rule"—that, balanced over the economic cycle, public debt should not increase for current expenditures.

[11] Contrary to the dictates of tax smoothing, moreover, many advanced economies were running up debt even though they faced rising health and other public expenditures related to aging populations.

[12] See https://clintonwhitehouse4.archives.gov/WH/New/html/Fri_Dec_29_151111_2000.html

average) there are two other sources of asymmetry that can generate a drift of debt ratios:

1. There could be an interaction between procyclical GDP forecasts and political economy considerations (discussed in section 3.A below) whereby the procyclical forecast bias during expansions is acted upon while the bias during recessions is (intentionally) ignored. If this is the case, the overall bias will be towards higher debt.
2. But there is also an asymmetry in terms of the way the economy reacts to procyclical fiscal policy as multipliers tend to be larger during recessions than during booms (Freedman et al. 2010; Auerbach and Gorodnichenko 2013; Jordà and Taylor 2016). As a result, the procyclical nature of fiscal policy will cause more damage to GDP during downturns than during expansions—especially when monetary policy is constrained by the zero-lower bound (and unable to stabilize output). There may even be hysteresis (i.e., permanent effects of cyclical shocks) such that the negative effects on GDP are permanent, validating the unfounded pessimistic expectations of governments. Such dynamics can lead to the perverse situation where a government engages in contractionary fiscal policy to reduce debt ratios but ends up instead with *higher* debt-to-GDP ratios. This is what the literature calls self-defeating fiscal consolidations (see Fatás 2018; or DeLong and Summers 2012). In this case, the resulting bias is again towards more debt despite the government's pessimistic view about GDP growth leading it to be overly conservative in its fiscal policy. Unlike in the previous cases, however, the solution here is for a more aggressive policy (larger deficits during crises) to avoid the negative effects on GDP.[13]

The accumulation of debt in times of crisis is not solely the result of standard Keynesian countercyclical policy but may also reflect government bailouts of the financial sector, which can result in as large or even larger increases in debt than stimulus spending (Campos et al. 2006; IMF 2015). This is important because, even if potential GDP forecasts are unbiased, estimates of debt will be too optimistic if they do not take account of the occasional support for the financial system during crises. Once these—hopefully rare—events happen, debt levels will be higher than expected. At that point, the logic of tax

[13] Of course, larger deficits are the optimal response assuming that there are no borrowing constraints by governments. In practice, some governments might see risk spreads and interest rates increase in a way that makes additional borrowing impossible.

smoothing that we have discussed earlier implies that debt levels will remain higher for a long time as the adjustment is optimally spread over many years (Ostry et al. 2015).[14]

C. Long-term investment and deficits

Investment projects financed by the government fall into the category of lumpy spending that, according to our tax-smoothing argument, should be financed by deficits. But unlike other forms of spending (i.e., government consumption), here the government is acquiring an asset that will deliver services in the future. This strengthens the argument for financing investment with debt.[15]

The welfare effects of debt-financed investment will depend on the social returns of the projects. Although this is also true for any other form of government spending, investment is typically assumed to deliver a social or even a financial return that justifies the spending, which might even generate the taxes necessary to repay the debt in the future. The greater the reliance on some nebulous social benefit in making the case for the investment, the greater the risk that the project turns out to be a white elephant. More generally, limited absorptive capacity could constrain the growth dividend of additional public investment projects, especially in periods of rapid acceleration of public investment (Presbitero 2018)—increasing the likelihood that the government later runs into debt-servicing difficulties (Case Study 3.1).

There are other instances where the government might also be acquiring assets and issuing debt to finance its purchases. One example is the issuance of external debt to finance the accumulation of foreign exchange reserves (which provide valuable FX liquidity during sudden stops or export shortfalls). Another example is financial sector bailouts that expand the government's balance sheet (Reinhart and Rogoff 2009; Laeven and Valencia 2013; Amaglobeli et al. 2017). When governments need to recapitalize the banking system, they acquire a financial asset (the equity stake in the bank), which they typically finance by issuing debt (in fact, the recapitalization often takes the form of a government bond). In this case we potentially have several arguments that justify these decisions. Stabilizing the financial sector can lead to a reduction

[14] Mauro (2011) surveys fiscal adjustment episodes by comparing ex-post outcomes with ex-ante plans.

[15] If the government were to buy this service from the private sector (i.e., the roads are built with private investment and then its services sold to the government) this would spread the spending naturally over time without the need for deficits.

in the severity of the decline of GDP. Not only will this be welfare enhancing, it will also raise tax revenues for the government. In addition, the assets acquired might be undervalued by the market because of fire sales or panic. In this case the assets could deliver a return that will partly or fully compensate the financial costs of issuing debt. A recent study on the fiscal costs of systemic banking crises over the period 1970–2011 shows that the median cost of direct government intervention in the banking sector amounted to about 7 percent of GDP (factoring in the indirect fiscal costs raises the impact of banking crises to 12 percent of GDP, International Monetary Fund (2015)).

A final case where deficits can be justified even if the government is not acquiring a physical or a financial asset is to support structural reforms. The political economy of structural reforms means that it is very difficult to find support for them in democratic environments (Fernandez and Rodrik 1991). This difficulty is the result of the uncertainty about who are the winners and the losers of those reforms. One way to ensure support would be through spending or tax measures that make those benefits immediate and reduce the potential uncertainty about the long-term benefits. In this case the government is investing in the necessary credibility needed to deliver structural reforms that will produce a social and even financial return in the long run (Banerji et al. 2017).

D. Asset management and government debt as safe asset

In Section 2.C, we emphasized the importance of looking at the asset side of the government's balance sheet to understand the existence of debt. The liability side may also be important. For instance, public debt may be issued not to meet the government's borrowing needs but to provide financial markets with risk-free instruments. Indeed, at the national level, government debt markets have often played a key role in developing nascent financial markets, including extending the yield curve to longer maturities and providing a benchmark.[16] International Monetary Fund (2012) highlights the overall benefits of safe assets. Abbas and Christensen (2010) show that moderate levels of non-inflationary government debt have a positive overall impact on economic growth. Gorton and Ordoñez (2013) analyze the benefits of government debt as a safe asset during crisis to show that within their

[16] The government debt market ("consols") was instrumental in the growth of Britain's financial markets, including the stock exchange, money markets, etc.) see Michie (2001: chs 1 and 2).

model "The decline in output during a crisis is lower to the extent that there are more government bonds outstanding."[17]

At the international level, there is a similar need for safe assets, and these assets are likely to be associated to one of the major reserve currencies (US dollar, euro, yen). In fact, the Global Financial Crisis—because of a combination of flight-to-safety and several sovereigns losing their AAA status—has resulted in a shortage of global safe assets with a variety of consequences (see Caballero et al. 2008; Gourinchas and Jeanne 2012; Brunnermeier et al. 2017, among others).

E. Dynamic inefficiency

Dynamic inefficiency—whereby the private sector cannot optimally provide vehicles to transfer wealth across generations—provides another potential rationale for government debt. In such an environment, issuing additional government debt is not only sustainable but also optimal (Blanchard 1985, 2019). For dynamic inefficiency to hold, it requires that the rate of return of an economy must be below its growth rate. Interest rates on government debt are often below GDP growth rates, but what matters is the rate of return on capital.[18] In a seminal study, Abel et al. (1989) provided strong evidence for six advanced economies that the criterion for dynamic inefficiency was not met. However, recent decades have seen substantial reductions in real interest rates on safe assets, which suggests it could be worth revisiting their findings. Geerolf (2017), concludes that dynamic inefficiency cannot be ruled out for several advanced economies. But whether the evidence is sufficiently compelling to warrant a clear policy recommendation in some of these countries remains an open question (Blanchard and Summers 2017; Blanchard 2019).

3. Bad Reasons to Issue Debt

Budget deficits, and the resulting accumulation of debt, can be optimal during recessions or in the presence of exceptional events such as war, natural

[17] Singapore is an interesting case of a government that has persistent surpluses but still issues debt to supply the financial system with a safe asset. Gorton and Ordoñez's (2013) results might not apply to countries that have difficulty financing large debts. As Jorda et al. (2011) show, countries that enter a financial crisis with high levels of debt have worse outcomes.

[18] While observed interest rates are often somewhat below GDP growth rates, if that were true on a persistent basis, then the government would effectively not face an intertemporal budget constraint.

disasters, or financial crises. Borrowing may also be justified by the need to finance large investment projects—though there is only limited evidence of a link between increases in public debt and surges of public investment (Figure 3.2).

A benevolent social planner would borrow up to the point at which the social marginal cost of an additional unit of debt (this includes principal, interest repayment, and any possible externality brought about by higher debt levels) equals the social return of an additional unit of debt-financed government expenditure. Overborrowing refers to a situation in which the government borrows more than is socially optimal. Yared (2019) suggests that the accumulation of public debt in recent decades is due to overborrowing driven by political distortions which leads to time inconsistent preferences and a bias towards present consumption.

A. Why do governments overborrow?

Just as it is wrong to compare the behavior of the government with that of a household—because the government is a large player and its borrowing decisions can have important spillovers, positive and negative, on the economy—it is also wrong to assume that policymakers always try to maximize social welfare. While it is reasonable to assume that the decision of a household head to borrow aims at maximizing the household's welfare, a policymaker's decision to contract public debt may be less benign. In order to understand why sovereigns overborrow, it is necessary to move from normative to positive theories of public debt (Alesina and Tabellini 1992).[19] The economics literature has emphasized four potential sources of excessive debt accumulation: (i) political budget cycles and rent seeking; (ii) intergenerational transfers; (iii) strategic manipulation; and (iv) common pool problems.

Political budget cycles and rent seeking

The literature on the political budget cycle suggests that politicians cut taxes and increase spending to increase the likelihood of being reelected. At the most basic level, the presence of a political budget cycle requires that voters suffer from "fiscal illusion." Only voters who do not fully understand the intertemporal nature of fiscal policy may be tempted to vote for politicians

[19] See Yared (2019), Alesina and Passalacqua (2016), and Battaglini (2010) for recent detailed reviews of the political economy of public debt.

who cut taxes or provide more public services without increasing taxes. In the traditional public choice literature, fiscal illusion is amplified by the asymmetric application of Keynesian stabilization policies, with policymakers happy to run budget deficits during recessions but less inclined to run surpluses in a period of rapid economic growth (Buchanan and Wagner 1977).

Political budget cycles models do not necessarily require voters to be irrational. For instance, Rogoff and Sibert (1988) develop a model in which the presence of fully rational but imperfectly informed individuals leads to a political business cycle because pre-electoral budget deficits are the only mechanism through which policymakers can signal their competence (the reason being that only competent politicians will be able to balance the budget after the election). Another implication of these types of models is that policymakers may engage in more visible, but not necessarily more efficient, types of public expenditure (Rogoff 1990).[20]

Political business cycle models implicitly assume that policymakers want to remain in power. This may be because of ego-rents or because, by staying in power, they can implement their favorite policies. It is, however, also possible that policymakers want to be in power to extract resources from the economy. Yared (2010) studies the behavior of politicians who try to extract rents and need to decide whether to extract a limited amount of resources in each period or extract everything they can in one period and then lose power. One of the implications of the model is that a high level of debt reduces the politicians' incentive to extract the maximum amount of rent and makes her more likely to behave as a social planner.

Intergenerational transfers

Individuals can leave positive bequests to their offspring but private negative bequests cannot be enforced by law. But individuals who would like to leave a negative bequest can use public debt to redistribute resources from future to current generations. Cukierman and Meltzer (1989) use an overlapping generations model to study an economy with two types of individuals: citizens who would like to leave a positive bequest to their children and constrained citizens who would like to leave a negative bequest. The first group only cares about public debt through its effect on the economy. Bequest-constrained

[20] A somewhat different strand of literature studies the link between public debt and political actions that go beyond voting. Using insights from behavioral economics Passarelli and Tabellini (2013) suggest that a perception of unfairness may lead to costly riots and that, in order to prevent such riots, the government will borrow more than what would be optimal if the citizens had full information on the available resources. In this case, excessive debt accumulation is a second best optimum.

individuals, instead, would like to issue more debt because this relaxes their constraint. In such a set-up, the level of debt depends on the bequest constraint faced by the median voter.[21]

Other models that focus on intergenerational transfers include Tabellini (1991) who develops a model with defaultable debt and wealthy and poor voters. In this setting, higher levels of debt (up to a point) create an incentive to repay the debt by linking intergenerational with intragenerational transfers. Song et al. (2012), instead, study a model in which the young and the old have different preferences for public goods and taxation is distortionary. In this setting, the level of debt is determined by these preferences (that can vary across countries) and the political power of the two groups. Jackson and Yariv (2015) show that if there are two groups of individuals and one group (the old) cares less about the future than the other group (the young) a government that aggregates the preferences of these two groups may suffer from a present bias. Yared (2019) shows that theory is consistent with the fact that there is a positive cross-country correlation between the growth rate of public debt and aging of the population. It is worth noting that the standard social planner model would predict the opposite correlation.

Strategic manipulation

On February 18, 1981 President Reagan described his program for economic recovery in a joint session of the US Congress. Among the topics discussed in his speech, there was the high level of public debt, which was approaching $1 trillion (this was total US Federal debt, Federal debt held by the public was about $770 million or 25 percent of GDP).[22] Eight years later, the US federal debt held by the public had surpassed $2.1 trillion (a 100 percent increase in real terms) and reached 39 percent of GDP.

Why would a conservative like President Reagan accumulate so much debt, and why are large and persistent primary surpluses often associated with left-of-the center governments (Eichengreen and Panizza 2016)? Persson and Svensson (1989) show that, in the presence of two parties with different preferences for spending and taxation, left-of-the-center parties (which prefer more public goods at the cost of higher taxes) may decide to run budget

[21] In the presence of imperfect capital markets individuals may prefer higher levels of debt to undo credit constraints faced by households. In such a set up a higher level of debt could be Pareto optimal and would not necessarily lead to any intergenerational transfer.

[22] The US national debt is approaching $1 trillion. A few weeks ago I called such a figure, a trillion dollars, incomprehensible, and I've been trying ever since to think of a way to illustrate how big a trillion really is. http://www.presidency.ucsb.edu/ws/index.php?pid=43425

surpluses so that the right-wing party will inherit a low level of debt and will not have a strong incentive to reduce public expenditure. Similarly, the right-wing party will increase the level of debt so that the left wing party will have to limit spending when in power.[23] In effect, debt is a state variable that the party in power can use strategically to constrain the actions of successor governments.[24]

While the model of Persson and Svensson (1989) shows how debt can be used to influence the actions of successive governments, it does not necessarily lead to excessive debt accumulation because deficits by right wing governments are canceled with surpluses by left wing governments. Alesina and Tabellini (1990) develop a model in which political parties have preferences for different types of public expenditure and accumulate debt in order to constrain the choices of future governments. In this setting, the level of debt depends on the likelihood of being reelected. Governments which are sure to stay in power behave like a social planner and issue no debt. However, governments with low probability of reappointment will overborrow.

Common pool

Common pool problems originate from the presence of externalities that lead to the private benefit of an additional unit of public expenditure being different from the social marginal cost of funding it (Olson 1965 and Ostrom 1990). The presence of concentrated interests amplifies the common pool problem. When policy actions benefit a certain group and are funded with a general tax, the relatively small group of people who benefit from the policy will have strong incentives to lobby in favor of the policy (for a classic analysis of the role of pressure groups see Buchanan and Tullock 1962). The much larger, but dispersed, group of actors that bears the cost of this action will have weaker incentives to act against it.

Common pool problems may explain why fiscal adjustments based on spending cuts, albeit desirable, are often hard to implement. For instance, Mauro and Villafuerte (2013) show that, while almost 90 percent of fiscal adjustment plans implemented by EU countries envisaged large spending cuts partly compensated by lower taxes, the data reveal that expenditure cuts often did

[23] Müller et al. (2016) develop a similar model with similar implications (in normal times a left-of-the center government issues less debt because it wants to be able to implement countercyclical policies in bad times), but in this case the incentive of the right-wing government to issue debt does not depend on its likelihood of remaining in power.

[24] Papers that emphasize the strategic role of debt include Persson and Svensson (1989), Aghion and Bolton (1990), Alesina and Tabellini, (1990), Tabellini and Alesina (1990) and Lizzeri (1999).

not materialize and that this lack of expenditure cuts either resulted in smaller budget surpluses (or outright deficits) or was compensated by temporary measures aimed at increasing revenues. This is clearly suboptimal as there is ample evidence that expenditure-based fiscal adjustments are preferable and more likely to be long-lasting than revenue-based fiscal adjustments.

The exact way in which the common pool problem manifests itself depends on the institutional setting. There is a large literature in political science (dating back to Weingast et al. 1981 and Baron and Ferejohn, 1989) that models common pool problems and pork-barrel spending in the US Congress. However, common pool problems also apply to situations in which the budget law is prepared by the government and then sent to the legislative body for approval. In such a setting, it is possible to think of a strategic interaction between the Ministry of Finance, which worries about the overall budget constraint, and the spending ministries which are subject to pressure from different interest groups (Alesina and Perotti 1996). Hierarchical rules in which the Ministry of Finance first decides the overall budget envelope and then the line ministries decide on the allocation may help reduce excessive spending (more on this in Section 3.B below).

Although common pool problems may lead to overspending, it does not necessarily follow that they will result in excessive budget deficits and debt accumulation as the higher spending could be matched by higher taxes. If property rights are not well defined, however, and each group fears that any residual government asset will be appropriated by the other group, each group may find it optimal to demand large transfers and push the government to its borrowing limit (Tornell and Lane 1999 and Velasco 2000).[25]

Political turnover amplifies common pool problems because if parties have different preferences for different types of public goods they will have an incentive to overspend on their favorite good when in power—the more so the lower the likelihood of being re-elected (Aguiar and Amador 2011). Empirically, Woo (2003) finds that budget deficits tend to be larger in countries characterized by deeper political cleavages and party fractionalization.

Common pool problems can also lead to overborrowing if legislators do not know whether they will be part of future governing coalitions. Battaglini and Coate (2008) show that adding uncertainty to a dynamic common pool model leads to two contrasting forces which unify the main findings of the normative literature on public debt (e.g., Barro 1979 and Aiyagari et al. 2002)

[25] Krogstrup and Wyplosz (2010) show that the presence of international externalities common pool problems can justify the presence of supranational debt ceilings.

with those of the positive literature that emphasizes the role of political failures (e.g., Alesina and Tabellini 1990). One the one hand, there is a self-insurance incentive: policymakers want to accumulate assets in order to insure against future shocks (as in Aiyagari et al. 2002). On the other hand, there is a political distortion: policymakers accumulate debt because they may not be part of future governing coalitions and higher levels of debt constrain the behavior of future policymakers as in the strategic models described above. When debt levels are low, political distortions dominate the self-insurance incentive and the government overborrows. As debt increases, the self-insurance motive becomes more important and fiscal policy becomes similar to the policy that would be chosen by a social planner (albeit with a higher equilibrium level of debt).[26]

B. Controlling overborrowing

The economics literature identifies three possible avenues to limit overborrowing. The first focuses on the electoral system, the second on fiscal rules, and the third on budgetary institutions.

Electoral systems

Battaglini (2010) shows that a simplified version of the model of dynamic electoral competition, discussed in Battaglini (2014), yields the unambiguous prediction that proportional electoral systems suffer from a deficit bias compared to majoritarian electoral systems. This prediction is in line with a large number of papers that show that democracies with a proportional electoral system accumulate more debt than democracies with a majoritarian system (e.g., Roubini and Sachs 1989; Grilli et al. 1991).[27]

A related literature that compares presidential and parliamentary democracies finds that the former tend to have smaller governments than the latter (and, within parliamentary democracies, majoritarian systems have smaller governments than proportional systems). The literature also find that in parliamentary democracies increases in government spending during recessions

[26] This literature tends to study the bargaining process within a legislature that includes representatives from different districts. There are also papers that focus on the electoral process and study how different parties choose a policy platform with the objective of winning an election (Battaglini 2014 and Battaglini and Coate, 2008).

[27] The degree of proportionality is usually measured with the size of electoral district (Taagepera and Shugart 1989; Lijphart 1994).

are less likely to be reversed during economic expansions (Persson and Tabellini 2003, 2004). If taxes remain constant over the business cycle, this behavior may lead to a ratchet effect and to a deficit bias in parliamentary democracies.

Fiscal rules

Fiscal rules aim at addressing the time inconsistency problem and limit debt accumulation by imposing an upper limit on budget deficits. A government that implements a fiscal rule trades off a constraint on its own action (something it does not like) against a constraint on successor governments (something it does like). Fiscal rules have become increasingly popular: while in the mid-1990s there were fewer than twenty countries with a national or international fiscal rule, there are now nearly 100 countries that adhere to some type of fiscal rule.[28]

The most extreme fiscal rule is the balanced-budget rule requiring zero deficits in every period. Such a rule may reduce welfare because it limits the government's ability to use countercyclical policies (or to smooth taxes).[29] A rule that aims at balancing the budget over the business cycle addresses this issue at the cost of being less transparent. Yared (2019) presents a detailed survey of these trade-offs by discussing the role of public information, the degree of enforcement (including the role of escape clauses), and the costs and benefits of rules based on specific targets (i.e., the total or primary deficits) vis-à-vis rules that concentrate on policy instruments (such as spending).

On empirics, there is a large literature on the effect of balanced-budget rules for subnational governments (especially US states see, for instance, Poterba 1996) as well as numerous studies of the fiscal rules adopted by many European countries. The results of this latter literature are mixed. On the one hand, Debrun et al. (2008) and Bergman et al. (2016) find that fiscal rules play a significant role in limiting budget deficits in European countries; on the other hand, Von Hagen (2006) suggests that the fiscal rules imposed by the Maastricht Treaty did not constrain the behavior of the largest countries in the euro area. The main challenge is to go beyond simple correlations and establish whether such rules have a causal effect on fiscal outcomes (Heinemann et al. 2018). Caselli and Wingender (2018), using an innovative

[28] For a recent discussion on fiscal rules see Eyraud et al. (2018). The IMF also maintains a comprehensive data set of fiscal rules. The data are available at: https://www.imf.org/external/datamapper/fiscalrules/map/map.htm

[29] Azzimonti et al. (2016) show that a balance budget rule is never welfare improving for economies with a positive level of debt.

identification strategy and a bunching estimation method, find that the Growth and Stability Pact has led to a bunching of fiscal deficits around the 3 percent Maastricht deficit ceiling.

Budgetary institutions

Preparing the budget is a complex exercise that involves several players within the government as well as interaction between the executive and the legislative. There is evidence that the institutions that regulate the preparation of the budget and guarantee its transparency have an impact on fiscal outcomes. In terms of budgetary process, hierarchical rules give more power to the ministry of finance, while collegial rules give more power to the spending ministries and allow the legislature to amend the budget. Hierarchical rules mitigate the common pool problem and are thus associated with smaller deficits and debt accumulation—albeit at the cost of democratic accountability (for surveys of the literature see Eichengreen et al. 2011; Hallerberg et al. 2009).

The transparency of the budget also matters. Rogoff and Sibert (1988) emphasize that imperfect information can lead to political business cycles and Milesi-Ferretti (1997) discusses how politicians who want to overborrow have incentives to window-dress their budget laws, even more so when the politicians are corrupt.[30] Standard strategies for manipulating the budget include keeping various items off-budget and adopting overoptimistic projections on either the state of the economy or on the effect of certain policies on tax revenues or expenditure.

Building on the intuition of Alesina and Tabellini (1990) and Rogoff and Sibert (1988), Beetsma et al. (2017) develop a model in which transparent budgets mitigate incentives to overborrow. This prediction is consistent with the empirical literature that finds that fiscal transparency is associated with lower levels of public debt (Alt and Lassen 2006; Alesina et al. 1999; Dabla-Norris et al. 2010).

C. The unexplained part of public debt

Thus far we have assumed that fiscal policy is the only driver of debt accumulation and that the stock of debt is equal to the sum of past deficits. However, debt accumulation is usually the sum of past deficits plus an unexplained

[30] Alesina and Cukierman (1990) show that politicians who favor policies which are different from those who would maximize their chances of reelection favor transparent budget procedures that do not reveal their preferences.

residual, sometimes referred to as the stock-flow reconciliation (or adjustment).[31] In some cases, this residual can be very large.

Consider the case of Uruguay. At the end of 2001, its net debt-to-GDP ratio was 35 percent (gross debt was 55 percent of GDP) and at the end of 2002 it had reached 76 percent (gross debt 106 percent). Yet over 2002 Uruguay's total budget deficit was only 3.7 percent of GDP; the increase in debt was 17 percentage points of GDP higher than the deficit. The case of Argentina is even more striking. At the end of 2001, Argentina's gross public debt was 49 percent of GDP, by the end of 2002 it had reached 152 percent of GDP while the 2002 public deficit was recorded at just above 2 percent. Debt grew 101 percentage points of GDP more than the deficit!

Campos et al. (2006) show that debt explosions—well beyond recorded deficits—are not limited to the few cases described above, and Cafiso (2012) shows that this phenomenon is not limited to emerging market countries and that the stock-flow reconciliation accounted for nearly one-third of public debt growth in EU countries over 2008–10.

The main drivers of this "unexplained" part of debt are balance sheet effects linked to the presence of foreign currency debt (Eichengreen et al. (2005); currency depreciations in the presence of dollar debt explain the debt explosions of Argentina and Uruguay mentioned above), banking crises (Amaglobeli et al. 2017), hidden deficits (and thus borrowing) driven by governments' incentives to misreport public expenditure (Weber (2012) shows that more transparent budgets are correlated with lower stock flow adjustments), and all sorts of contingent liabilities linked to implicit subsidies and public guarantees.

While some of these fiscal risks are unavoidable, policymakers could implement policies aimed at mitigating them. For instance, a safer (from the point of view of the borrower) debt structure could mitigate some of the risks linked to balance sheet effects, and contingent debt instruments, such as GDP indexed bonds, could reduce fiscal costs at a time of crisis (Borensztein and Mauro 2004). The question is why these safer debt instruments are so seldom used.

Part of the answer is that local currency or indexed debt tends to be more expensive than what a fair insurance would predict—both because of market failures and because of the incentives (highlighted by Calvo 1988 and Tirole 2003) associated with local currency debt. However, political

[31] The name derives from the fact that this residual entity reconciles the deficit, which is a flow variable with debt, which is a stock variable.

failures also play a role as domestic currency debt and contingent instruments more generally work as an insurance policy: their cost must be paid upfront but their payoff may not occur until years later (or never occur). As such they tend to be poorly understood by the public and may not be desirable for policymakers who may be afraid to pay the cost of an instrument that might end up benefiting their successors (for a detailed discussion see Borensztein et al. 2006).

4. Debt, Growth, and Investment

Regardless of the motives to borrow, high levels of government debt can have adverse effects on the economy, limiting policymakers' capacity to run countercyclical fiscal policy, crowding out private sector investment, tightening credit constraints, and creating the expectation of higher future distortionary taxation. Conversely, public borrowing—even if it results in a higher debt ratio—can be good for growth if used for productive investment or demand stimulus during downturns (see above, Section 2.C). Nevertheless, the increase of investment spending must be balanced against considerations of debt sustainability—a particularly pressing issue in many developing countries that need large investments to realize the 2030 Sustainable Development Agenda (see Case Study 3.1).[32]

If Ricardian Equivalence does not hold, the decrease in public saving associated with debt accumulation will not be fully compensated by higher private saving and will lead to a lower stock of capital and thus higher interest rates and lower economic growth (Diamond 1965; Blanchard 1985). This is the classic crowding-out effect, which can also be obtained within a simple IS-LM model. Using the back-of-the envelope calculations of Elmendorf and Mankiw (1999) and Panizza and Presbitero (2013) show that this effect is not quantitatively large.[33] The crowding-out effect of public debt could, however,

[32] However, the empirical literature on the growth effects of public investment is not conclusive. While there is evidence of a positive growth effect of debt-financed public investment in advanced economies (Abiad et al. 2016), a study of several episodes of public investment booms casts doubts on this positive narrative and suggests that the growth impact could be very limited, at the cost of larger ex-post public debts (Werner 2014).

[33] By assuming that an annual real GDP growth is 3 percent and a convergence speed of 2 percent, Panizza and Presbitero (2013) show that the steady-state change in output computed by Elmendorf and Mankiw (1999) implies that increasing debt by 50 percent of GDP would reduce annual GDP growth by approximately 10 basis points in the first 20 years. In a three-asset setting, Friedman (1978) argues that higher government debt can "crowd-in" private capital accumulation, depending upon the substitutability between assets in financial portfolios.

become large if government debt results in tightening credit conditions faced by private firms (Broner et al. 2014). The recent evidence of the European sovereign debt crisis has shown that the expansion of the share of government debt held by the banking sector in times of crisis crowds out private sector lending (Altavilla et al. 2017; Becker and Ivashina 2018).

High public debt can also have a negative effect on economic activity as investors anticipate lower real returns in the future—whether as a result of confiscation, inflation, or higher taxes (Cochrane 2011). Amidst sustainability concerns, moreover, rising debt can result in widening spreads, which are then transmitted to the rest of the economy, raising the cost of borrowing for the private sector (Codogno et al. 2003; Laubach 2009; Baum et al. 2013).[34]

Governments generally react to increasing public debt with austerity measures, running smaller deficits or larger surpluses (Bohn 1998; Mendoza and Ostry 2008; Ghosh et al. 2013; Mauro et al. 2015). In this respect, high levels of public debt may also have a negative impact on growth as they could limit a country's ability to conduct countercyclical policies and, possibly, lead to self-defeating austerity policies thus increasing output volatility and reducing economic growth. As the relationship between the level of debt and the ability to conduct countercyclical policies is also dependent on the composition of public debt (Hausmann and Panizza 2011; De Grauwe 2011), countries with different debt structures may start facing problems at very different levels of debt.

A. What do the data say?

The rapid increase in public debt in the aftermath of the Global Financial Crisis sparked a large empirical literature on the growth effects of public debt. An influential paper by Reinhart and Rogoff (2010a) uses data for twenty advanced economies over 1946–2009 to plot average GDP growth for different levels of debt and shows that average and median growth is substantially lower when public debt surpasses 90 percent of GDP. This finding is robust to using more recent data from the newly available Global Debt Dataset (Mbaye et al. 2018), with average (median) growth declining from 3.7 percent when the debt-to-GDP ratio is less than 30 percent, to 2.6 (2.7) percent when the debt ratio is between 30 and 60 percent, and to 1.2 (1.6) percent when debt

[34] Indeed, the debt overhang literature (Krugman 1988; Sachs 1989; Aguiar et al. 2009) suggests that there is a level of debt at which these growth effects are so large that debt relief would benefit both debtors and creditors.

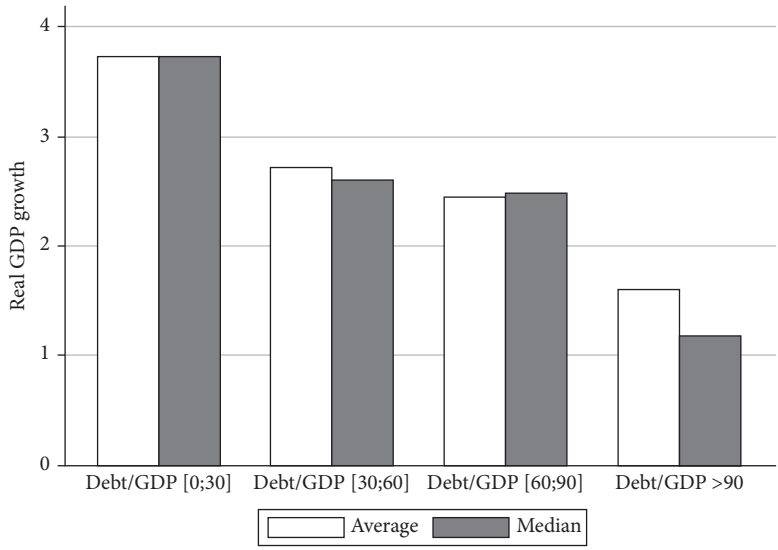

Figure 3.3 Government debt and growth, selected advanced economies; 1960–2016

Notes: The sample includes twenty advanced economies as in Reinhart and Rogoff (2010a, Figure 2): Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, the United Kingdom, and the United States. Data refer to central government debt, apart from the Netherlands, for which general government data have been used, because of data availability.

Source: Global Debt Dataset (Mbaye et al., 2018) and World Economic Outlook.

surpasses 90 percent of GDP (Figure 3.3). However, these differences are smaller when looking at a large sample of 119 low- and middle-income countries, where average growth declines from 4.4 percent for low-debt countries to 2.6 percent in high-debt (above 90 percent) countries (Figure 3.4).

Reinhart and Rogoff's (2010a) article was followed by a large number of papers aimed at assessing whether the correlation between debt and growth was robust to controlling for other variables in a formal regression set-up, and to instrumenting public debt to assess its causal effect on economic growth. Another set of papers focuses on non-linearities allowing for a non-arbitrary debt bracket.

By and large, there is strong evidence that public debt is negatively correlated with future economic growth. We corroborate the negative correlation between debt and growth by plotting current debt level and future growth and showing that, controlling for year- and country-fixed effects, there is a strong negative correlation between the debt-to-GDP ratio in year t and real GDP growth between t and $t+5$ (Figure 3.5). This negative correlation does not necessarily imply that high levels of debt *cause* lower growth. Indeed, the negative

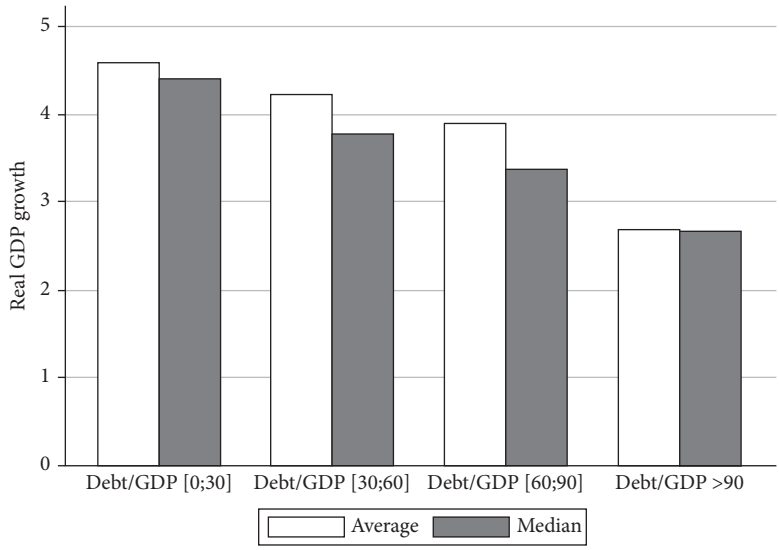

Figure 3.4 Government debt and growth, low- and middle-income countries; 1960–2016

Notes: Data refer to central government debt. The sample includes 131 low- and middle-income countries. Data refer to central government debt.

Source: Global Debt Dataset (Mbaye et al., 2018), World Development Indicators and World Economic Outlook.

correlation between debt and growth could be driven by unobservable omitted variables that are jointly correlated with debt and growth or by reverse causality. Slow economic growth, in fact, is an important factor explaining the rise of the public debt-to-GDP ratio (see, for instance, the Italian experience discussed in Case Study 3.2). This is not only because of the direct effect of growth on the denominator of the debt ratio, but also because the primary surplus depends on economic growth. In fact, absent any policy measure, low growth acts as a constraint to public revenues, while expenditures increase in line with inflation, leading to a larger deficit and a rising debt (Mauro and Zilinsky 2016).[35] There is also a third factor that goes beyond the mechanical impact of growth on GDP and revenues. Mauro et al. (2015) show that permanent growth slow down is often associated with increasing debt because policymakers often confuse changes in trend growth with temporary slowdowns and try to use expansionary fiscal policy to address a structural problem.

[35] Mauro and Zilinsky (2016) show that the difference in growth rates between Ireland and Italy after the global crisis is the main driver of the diverging debt trajectories. More generally, they show that economic growth is a key factor to explain the debt evolution in most advanced economies since the 1950s.

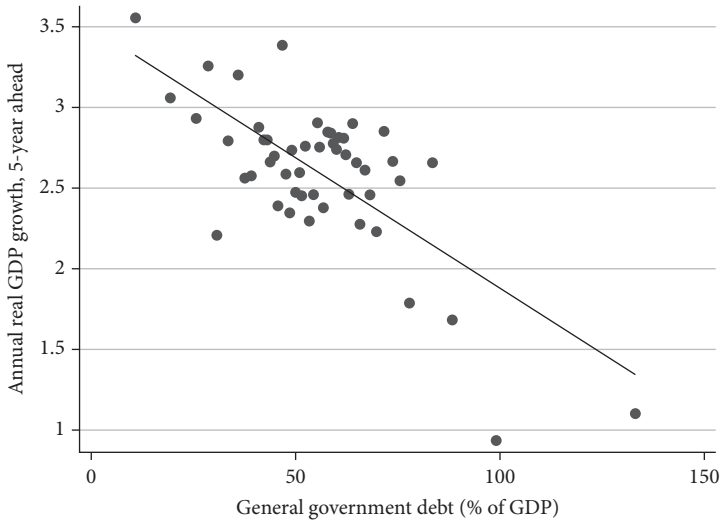

Figure 3.5 Government debt and subsequent GDP growth, selected advanced economies; 1960–2016

Notes: A regression of the annual real GDP growth between *t+5* and *t* against the ratio of general government debt over GDP at time *t*, controlling for year and country fixed effects, gives a coefficient on the debt variable of –0.016 (p-value of 0.001), meaning that 10% higher debt-to-GDP ratios are associated with 0.2% lower future growth over five years. To generate the binned scatterplot, starting from the sample of nineteen OECD economies (data on general government for New Zealand are not available), the annual real GDP growth between *t+5* and *t* (y-axis) and the ratio of general government debt over GDP at time t (x-axis) are regressed against year and country fixed effects. Then, the x-residuals are grouped into fifty equal-sized bins, then the chart plots, for each bin, the mean of the annual real GDP growth between *t+5* and *t*, within each bin, holding the controls constant. The line is the linear fit of the OLS regression of the y-residuals on the x-residuals. The number of observations is 923.

Source: Global Debt Dataset (Mbaye et al., 2018) and World Economic Outlook.

Case Study 3.2 The evolution of Italian public debt after World War II

Italy has the fourth largest stock of public debt in the world, the second highest debt-to-GDP ratio in the G7 group of advanced economies, and the highest debt service ratio in the G7.

Italian public debt stood at 74 percent of GDP at the end of World War II in 1945, but dropped to 24 percent of GDP by 1947.[1] The main driver of this rapid debt reduction was high inflation (20 percent in 1946 and 62 percent in 1947) and the corresponding negative real interest rates (the

[1] All the data refer to gross debt. In Italy the difference between gross and net debt is not very large. For instance, WEO data for 2016 report a gross debt of 131% of GDP and a net debt of 119% of GDP.

Continued

Case Study 3.2 *Continued*

stock of debt still carried single digit interest rates). From this point on, it is possible to identify seven phases in the behavior of the Italian public debt: (i) a moderate increase (to 34 percent of GDP) during the postwar reconstruction period; (ii) a slight decrease (to 27 percent of GDP) over 1955–64; (iii) a rapid increase to (to 56 percent of GDP) over 1964–75; (iv) a period of relative stability in the second half of the 1970s; (v) a rapid increase over 1980–94, with debt peaking at 120 percent of GDP in 1994; (vi) a consolidation phase over 1995–2007, with debt bottoming at 103 percent of GDP in 2004; and (vii) a rapid increase after 2008, with debt surpassing 130 percent of GDP in 2016.

These different phases can be explained by different economic and political events. The increase in debt during the reconstruction period was driven by massive public investment, which was needed to rebuild and modernize Italian infrastructures (by the early 1960s, public investment still accounted for more than 20 percent of total public expenditure). Over 1955–64, debt reduction was driven by a combination of rapid economic growth and small overall budget deficits (composed of small primary surpluses that almost fully balanced interest payments; in the postwar period Italy never ran an overall budget surplus). In the second half of the 1960s, Italy started running increasingly larger primary deficits, hovering at around 2 percent GDP in the second half of the 1960s and going well above 4 percent of GDP in the 1970s. These large budget deficits are partly explained by the political economy theories illustrated in Section 3.3 as they were associated to a period of often short-lived coalition governments, a strengthening of the opposition Communist Party (which in late 1960s and early 1980s was receiving more than 30 percent of Italian votes), and a period of political and labor tensions that culminated in the "Hot Autumn" of 1969–70.[2]

The interesting fact is that while primary deficits led to a debt explosion in the second half of the 1960s and early 1970s, even larger primary deficits had almost no impact on debt in the second half of the 1970s. The explanation has to do with the conduct of monetary policy which, by the late 1970s was mostly driven by the need to finance growing deficits. Debt monetization

[2] The term "Hot Autumn" denotes a series of large strikes in the main industrial cities of Northern Italy.

led to high inflation and, with the help of some financial repression, negative real interest rates, which kept debt under control over 1973–80.[3]

The situation changed when, in 1981, the "divorce" between the Bank of Italy and the Treasury prevented the central bank from monetizing the deficit. The divorce led to a rapid reduction of inflation and a sudden increase of real interest rates.[4] The conquest of fiscal dominance, however, was not followed by fiscal reforms, and Italy kept running primary deficits until 1991. The combination of prudent monetary policy and large primary deficits led to the explosion of public debt over the 1980s and early 1990s. Starting from 1991, Italy began running primary surpluses. However, with high real interest rates debt kept growing until 1994. In the first half of the 1990s, interest payments absorbed more than 20 percent of total public outlays (in 1992 Italy ran a primary surplus of 1.8 percent of GDP but the overall budget deficit was above 10 percent of GDP).

Starting in 1994, a combination of substantial primary surpluses and decreasing real rates lead to a process of debt reduction. This process of fiscal consolidation, driven by higher revenues (revenues reached 47 percent of GDP in 1997, partly due to special "below the line" operations) and unchanged primary expenditure (at about 41 percent of GDP), slowed down over 2000–05. Primary surpluses went from the 4–6 percent range of the late 1990s to a 1–3 percent range; as a consequence, the debt-to-GDP ratio started growing again in 2005–06. There was a brief moment of consolidation in 2007 immediately interrupted by the explosion of the global financial crisis in 2008/2009. The first half of the new millennium was thus a missed opportunity for further reducing public debt, and Italy entered the global financial crisis with a level of debt well above 100 percent of GDP.

The post global financial crisis period was characterized by fiscal restraint (with Italy running a small primary deficit in 2009 and 2010 and primary surpluses well above its European counterparts in the remaining years), but with a contracting economy, which lead to a rapid increase in the debt-to-GDP ratio. During the crisis, the high level of debt associated with past fiscal profligacy constrained Italy's policy response and possibly contributed to magnifying the crisis. In this sense, the high level of inherited debt may have a led to suboptimal macroeconomic policies with a negative effect on Italy's GDP growth during the crisis—and, correspondingly, worsening debt dynamics.

[3] Consumer price inflation was almost always above 15% over 1974–82 (it was 12% in 1978) and peaked at 21% in 1980.

[4] Inflation went from 21% in 1980, to 9% in 1985, 5% in 1987, and 2% in 1997.

Thus, establishing whether debt has a *causal* effect on growth requires an instrumental variable or a natural experiment that allows the researcher to isolate exogenous changes in public debt. In the presence of persistent variables like the debt-to-GDP ratio, the standard approach of using past values of the variables of interest as instruments does not solve the identification problem (Reed 2015; Bellemare et al. 2017). Panizza and Presbitero (2014) propose a strategy that uses valuation effects brought about by the presence of foreign currency debt as an exogenous driver of the change in public debt and find that public debt has no effect on future growth. One problem with this approach is that in their sample of advanced economies the share of foreign currency-denominated debt is relatively small, and hence the instrument is not very strong. Another way to achieve identification is to move from macro to micro data. Huang et al. (2017, 2018) match firm-level balance sheets with data on either public debt across a sample of sixty-nine countries or local government debt across 270 Chinese cities to show that government debt tightens financing constraints for private sector manufacturing firms. In a similar vein, Croce et al. (2019) identify a different channel through which debt can affect productivity and growth showing that, in the United States, higher government debt increases the cost of capital and has a negative effect on investment by R&D-intensive firms. There is, however, a trade-off between identification and the ability to assess the macroeconomic effects of debt accumulation. While firm-level analysis allows precisely testing one channel through which debt may have a negative effect on growth, it "hides" the potential macroeconomic links between debt and growth. For instance, it is possible that higher levels of debt increase investment for all industries and firms considered by Huang et al. (2017, 2018), but that investment increases less for credit-constrained firms.

Besides studying the average correlation between debt and growth, the economics literature also seeks to identify possible non-linearities and threshold effects. Some analyses indicate that the debt trajectory can have more important consequences for economic growth than the level of debt-to-GDP itself (Pescatori et al. 2014; Chudik et al. 2017), in line with recent evidence on how public debt could affect debt sustainability and market access (Bassanetti et al. 2019).

Moving to the presence of debt thresholds, the notion that there is a non-linearity in the debt–growth relationship and that this non-linearity is at a specific value of the debt-to-GDP ratio—often taken to be 90 percent—has become popular.[36] However, assessing non-linearities is complicated by

[36] Reinhart and Rogoff (2010a, 2010b) did not explicitly suggest the presence of discontinuities when debt reaches a certain level. Their view is that they "do not pretend to argue that growth will be

lack of statistical power due to the limited number of observations above the relevant threshold, and it is possible that the results of the literature are driven by the imposition of some parametric approach and on a few outliers.[37] Moreover, this literature imposes common coefficients and thresholds across countries, while the data suggest that there is substantial heterogeneity especially when looking at larger samples, which pool together developing and emerging economies as well (Eberhardt and Presbitero 2015; Chudik et al. 2017). Consider, for instance, Figure 3.6, which plots the outcome of a

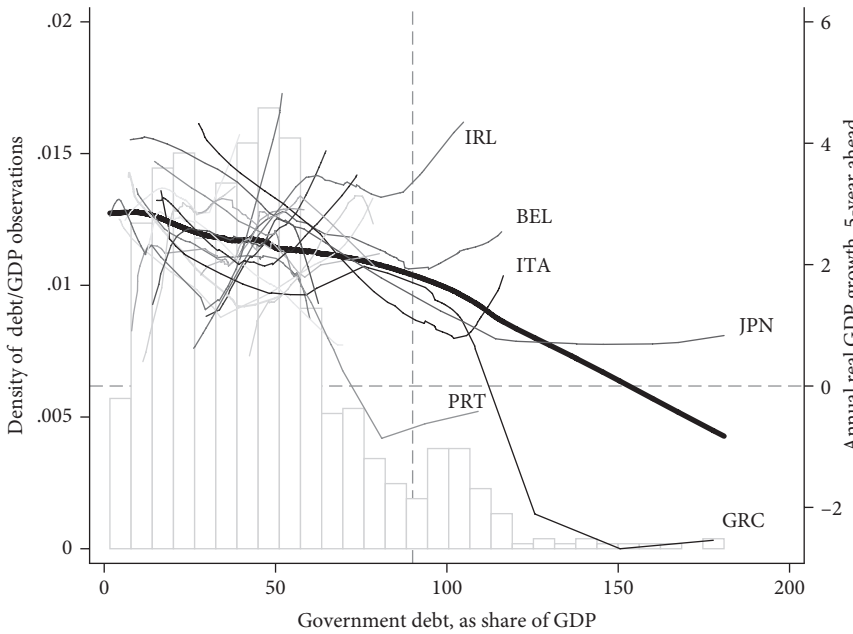

Figure 3.6 Non-linearities and heterogeneity in the debt–growth relationship

Notes: The solid black line plots smoothed values of the locally weighted regression of the annual real GDP growth between *t+5* and *t* against the ratio of general government debt over GDP at time *t* for the whole sample. The thin lines plot the same smoothed values for single countries. The histogram shows the density distribution of the ratio of general government debt over GDP (x-axis). The sample includes twenty advanced economies as in Reinhart and Rogoff (2010a, Figure 2): Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, the United Kingdom, and the United States. Data refer to central government debt, apart from the Netherlands, for which general government data have been used, because of data availability.

Source: Global Debt Dataset (Mbaye et al., 2018) and World Economic Outlook.

normal at 89% and subpar (about 1% lower) at 91% debt/GDP any more than a car crash is unlikely at 54mph and near certain at 56mph."

[37] The evidence of the actual presence of a common debt threshold in these studies is weak. See Panizza and Presbitero (2013) for an overview and Ash et al. (2017) for a replication of some of the

non-parametric regression based on a sample of twenty advanced economies over the period 1960–2016 and shows that (i) the average negative correlation between debt and future (5 years ahead) growth hides a large degree of heterogeneity across countries, and (ii) while the relationship between debt and growth is nonlinear there is no common threshold beyond which an increase in debt is associated with a growth slowdown.

One reason for the presence of country-specific thresholds is that the level at which public debt becomes "too high" must depend on country characteristics. For example, in the context of sovereign default, Reinhart et al. (2003) classified countries into clubs and "debt intolerance" regions, which depend not only on borrowers' debt levels, but also on their credit and inflation history. Likewise, Ghosh et al. (2013), in calculating fiscal space, find that governments' debt-sustainability thresholds depend on their historical track record of fiscal adjustment in response to rising public debt. Alternatively, Eichengreen et al. (2005) emphasized the role of debt composition. In the debt and growth literature, Kourtellos et al. (2013) explicitly modeled the possibility of different regimes depending on a large set of country characteristics and find that only when institutions are below a certain level does higher debt translate into lower GDP growth. As countries with poor institutions also have higher debt levels, these results provide a mechanism to interpret (and are consistent with) the general finding of a negative relationship between debt and growth. Specifically, countries with low-quality institutions may be more inclined to the political budget cycle and less able to control overborrowing. In addition, those countries could have a higher propensity to finance government consumption rather than productive investment, leading to higher debt and lower growth.

Other authors have looked at the dynamics between debt and growth from an historical perspective. Esteve and Tamarit (2018) focus on the Spanish economy for the period 1851–2013 and find some support for a negative relationship between public debt and growth, but no clear evidence of a debt threshold. Balassone et al. (2013) consider the experience of Italy since its unification in 1861 and find that when debt exceeds 100 percent of

most widely cited studies. For instance, Woo and Kumar (2015) run a simple growth model interacting the debt-to-GDP variable with three dummies for ratios: (i) below 30 percent, (ii) between 30 and 90 percent, and (iii) above 90 percent. In 2 (out of 4) specifications of their Table 5, they find that the coefficient of the debt ratio is negative and significant when larger than 90 percent, but this coefficient is lower than (equal to) that for debt between 30 and 90 percent in the OLS (GMM) estimates. In other words, they cannot test that the *correlation* between debt and growth is statistically higher when debt is larger than 90 percent of GDP. Checherita-Westphal and Rother (2012, Table 3), instead, run a quadratic model and report the confidence intervals of the turning point, which is 49–119 percent of GDP.

GDP its negative effect on growth becomes stronger. However, Eberhardt (2019) challenges this conclusion and, adopting a more flexible framework and data over more than two centuries for Great Britain, Japan, Sweden, and the United States, finds no evidence for any long-run non-linear relationship between debt and growth.

Overall, our reading of the empirical literature is that, at least in advanced economies, there is a negative correlation between public debt and subsequent economic growth but no convincing evidence of *causality*: high debt and low growth may just reflect a weak macroeconomic framework, which is driving both aggregates. A major complicating factor in any empirical analysis is that not all debts are equal: factors such as what the debt was used for, who holds it, its currency composition and maturity are all likely to affect any impact on future growth. While it is widely recognized that these should be taken into consideration (Eberhardt and Presbitero 2015; Chudik et al. 2017), the lack of data and of consistent cross-country definitions precludes most empirical analysis (Panizza and Presbitero 2013). Ultimately, the relationship between the accumulation of public debt and subsequent growth performance is likely to be highly complex, two-way, and country-specific. Some six years after Reinhart et al. (2012: 80) we still agree with them that the "endogeneity conundrum has not been fully resolved".

Finally, it bears emphasizing that even if is true (in a causal sense) that "debt is bad for growth" it does not necessarily follow that governments should pay down the existing debt (Ostry et al. 2015). In terms of social welfare (i.e., the distortionary cost), it may be more costly to pay down the debt than to live with it (provided, of course, the government does not face a funding crisis). In steady-state, this result is follows directly from tax smoothing. Unless taxes are set to just service the debt indefinitely, they will either have to be increased in order to maintain sustainability against a growing debt, or they will have to be decreased once the debt has been repaid—either violates the principle of smoothing taxes to minimize the distortionary costs. Interestingly, the result also holds out of steady-state—at least for an important class of utility functions (i.e., iso-elastic). This is because, even though the presence of distortionary taxes implies wedges between private and social marginal products and rates of substitution, the market interest rate equals the discount rate of a benevolent government (i.e., that seeks to maximize the representative agent's utility).[38] The government can choose to pay down $1 of domestic debt today at a certain distortionary cost.

[38] See Chari et al. (1994) for an analogous result.

Or it can wait till tomorrow, when the debt and the cost will have grown by (1+r), the market interest rate. But the government discounts the future at precisely (1+r), so it is indifferent between paying down the debt today or tomorrow. Since the same argument holds across all periods, the steady-state result—that it is optimal to just live with the inherited debt—obtains even out of steady state.

5. Concluding Remarks

Governments issue debt for a variety of reasons—both good and bad. Among the good reasons are intertemporal tax smoothing, fiscal stimulus during economic downturns, and asset management, including providing financial markets with safe assets. While such motives can explain some of the increases in public debt—in particular, after wars or major financial crises—they cannot plausibly account for all of the observed changes. The correlation between public investment and public borrowing—supposedly a major non-wartime motivation for issuing debt—is surprisingly weak. Indeed, the behavior of governments is sometimes quite at odds with these theories. A notable example is the build-up of public debt in many advanced economies during the early 2000s, when the world economy was booming, and the looming prospect of aging-related costs should have spurred public saving.

Counter-cyclical fiscal policies with implementation delays and forecast biases might be part of the explanation for the upward trend in public debt in many advanced economies. But a full accounting needs to go beyond purely economic rationales and consider social, political, and institutional factors that might be at play. Politicians pursuing their own self-interest and seeking to maximize their chances of re-election may engage in a political business cycle that results in debt rising over time. Strategic manipulation whereby the party in power seeks to circumscribe its (possible) successor's ability to spend public funds by deliberately running up public debt will likewise result in a positive debt bias. And common pool problems, which result in the private benefit of an additional unit of spending exceeding the social marginal cost of funding this extra unit of expenditure, provide a third political economy explanation. More generally, where the "safety-valve" of inflation is unavailable (e.g., under a currency board regime or membership in a monetary union), competing demands by different socioeconomic groups is often (temporarily) resolved through rising public debt.

But why does overborrowing matter? And what can be done about it? Other chapters in this volume explore some of the consequences of excessive

government borrowing—including debt sustainability problems and possible crises. Even in the absence of crises, however, public debt can be costly. In welfare terms, the cost of public debt is the present discounted value of distortions associated with the taxes necessary to service that debt. Empirically, there is a negative relationship between public debt and output growth. The jury is still out on whether that relationship is causal—higher levels of public debt impeding growth—and in reality, the answer must depend on what the debt was used to finance, how it is expected to be repaid or serviced, and a host of other country-specific factors.

As to measures that democracies can take to limit overborrowing, the literature has identified three key avenues: electoral systems, fiscal rules, and budgetary institutions. While their effectiveness will depend on country circumstances, all imply some trade-off between the flexibility to respond to shocks and to issue debt for good economic reasons and the need to discipline policymakers from borrowing excessively.

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